

- $F_{clk} = 40 \text{ MHz}$

$$T_{clk} = \frac{1}{F_{clk}} = \frac{1}{40 \times 10^6} = 25 \text{ ns}$$

- $T_{clk_{FF1}} = 2 + 2 + 2 = 6 \text{ ns}$

$$T_{clk_{FF2}} = 2 + 2 + 2.5 + 2.5 = 9 \text{ ns}$$

- $T_{skew} = T_{clk_{FF2}} - T_{clk_{FF1}} = 9 - 6 = 3 \text{ ns}$

↳ positive skew

- clock jitter $\begin{cases} \rightarrow \text{For setup} = -2 \text{ ns} \\ \rightarrow \text{For hold} = +2 \text{ ns} \end{cases}$

* Setup :

Maximum delay

• path (1) :

$$FF_1 \rightarrow 5 \text{ ns} \rightarrow \text{Comb 1} \rightarrow FF_2$$

- $T_{data} = T_{cq_{f1}} + T_{wire} + T_{comb1}$

$$= 6 + 5 + 12 = 23 \text{ ns}$$

- $T_{arrival} = T_{clk_{FF1}} + T_{data} = 6 + 23 = 29 \text{ ns}$

$$\begin{aligned}
 \bullet T_{req} &= T_{period} + T_{clk_FF2} + T_{su2} + T_{Jitter} \\
 &= 25 + 9 + 4 - 2 \\
 &= 28 \text{ ns}
 \end{aligned}$$

$$\bullet \text{slack} = T_{req} - T_{arrival} = 28 - 29 = -1 \text{ ns}$$

↳ Negative slack $\rightarrow$ Setup violation

• path (2):

$$FF_1 \rightarrow 2 \text{ ns} \rightarrow \text{Comb2} \rightarrow FF_2$$

$$\begin{aligned}
 \bullet T_{data} &= T_{cq1} + T_{wire} + T_{comb2} \\
 &= 6 + 2 + 13 = 21 \text{ ns}
 \end{aligned}$$

$$\bullet T_{arrival} = T_{clk_FF1} + T_{data} = 6 + 21 = 27 \text{ ns}$$

$$\bullet T_{req} = 28 \text{ ns}$$

$$\bullet \text{slack} = 28 - 27 = 1 \text{ ns}$$

↳ Positive slack $\rightarrow$ No Setup violation

* Hold:

Minimum delay

• path (1):

$$T_{data} = T_{cq_1} + T_{wire} + T_{comb_1}$$

$$= 2 + 5 + 2 = 9 \text{ ns}$$

$$T_{arrival} = T_{clk_{FF_1}} + T_{data} = 6 + 9 = 15 \text{ ns}$$

$$T_{req} = T_{clk_2} + T_{hold} + T_{Jitter}$$
$$= 9 + 3 + 2 = 14 \text{ ns}$$

$$Slack = T_{arrival} - T_{req} = 15 - 14 = 1 \text{ ns}$$

→ positive slack

• path (2):

$$T_{data} = T_{cq_1} + T_{wire} + T_{comb_2}$$

$$= 2 + 2 + 1 = 5 \text{ ns}$$

$$T_{arrival} = T_{clk_1} + T_{data} = 6 + 5 = 11 \text{ ns}$$

$$T_{req} = T_{clk_2} + T_{hold} + T_{Jitter} = 14 \text{ ns}$$

$$Slack = T_{arrival} - T_{req} = 11 - 14 = -3 \text{ ns}$$

→ Negative slack → hold violation